PEER REVIEW

MScFE Capstone Project

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Peer Review

Paper reviewed: Ejike, Uchenna Splendor, Israel Olanrewaju Odeajo and Goodness Chibueze Kalu. "Path-Robust Risk Controls for Intraday Mean Reversion in NSE Equities." Project Proposal, MScFE 690 Capstone, Student Group 17838, 10 October 2026.

Reviewers: Ikemefula Solomon Oriaku and Zione Esther Kamtambe. Date of review: 17 October 2026.

The proposal tests three exit policies on one mean-reversion entry rule over minute data (2021 to 2023) in five Nifty constituents, with a frozen 20-session pilot and an expanded development run of 443 sessions. The work is unusually transparent and explicit in stating its data, accounting and inferential limitations, and its main empirical claims are tempered. We will comment on the match between the title and the evidence, the economic and statistical significance of the main comparison, a few aspects of design that restrict the lessons that can be drawn from the text, and the clarity of the writing. The page references are to the submitted PDF. We have also re-run the supplied engine on the pilot inputs and compared its output with the reported tables, and read the research notebook cell by cell against the written protocol, so the factual statements below are based on these checks.

1. Objectives (The Major Theme)

The research question is clearly laid out in the Introduction (Ejike et al. 1): which exit strategy (stop-only or a combination of stop-loss and profit-target) provides greater net returns or lower downside risk than fixed-time exits when the entries, position sizes and expiry are held identical? A combined-minus-time mean daily return is the primary estimand that is declared in advance of the results, and the matched design is appropriate to this estimand. The scope is also well limited in light of the earlier feedback summarised in Table 1.

The biggest problem is that there is an uneven distribution of the objective across three aims which are not ordered: an exit-policy comparison, an audit of candle ambiguity, and evaluation of the reliability of block-bootstrap inference. The evidence says that there was no episode whatsoever of any ambiguity, despite the promise of path-robust controls (0 of 637 and 0 of 662, p. 11). The path-robustness feature is therefore never used on the data, while the title is exaggerated based on the results. Also, the abstract does not include a research question, and starts with a method and not a motivation.

Questions for the authors. Which of the three aims is the major contribution and which are supporting diagnostics? Would a title such as “An Audit of Exit Rules under Explicit Data and Inference Limits” better capture the true focus of the study? What result, in either direction, would count as an answer to the research question?

2. Theory (Assumptions, Weaknesses or Gaps)

The theoretical framework (pp. 2 to 3) is presented in a sensible way, based upon the reversal under costs, the influence of stops and targets, and observation and evaluation problems. Kaminski and Lo are used appropriately as the primary source to indicate that a stop cannot add

value when returns are independent and the proposal acknowledges that their environment is different from that of the setting of Kaminski and Lo's portfolio overlay (Ejike et al., 13 to 14).

There are three suppositions that need to be examined. First, it is tacitly assumed that there is some benefit to the entry rule, but at the reference cost, all three policies make a negative expected profit on average (p. 9). If the entry has no gross edge, then it is already predicted by the theory the authors mention that no exit rule can help, and the exit comparison is a comparison of three paths to loss. The gross 15-minute return of the signal in the proposal is not mentioned; thus, it cannot be determined whether the exits are being compared on a strategy that is worth improving. The gross time-only return from the pilot means and the 15 basis point round-trip cost is about minus 2.7 basis points per trade under start labels and about plus 7.6 basis points per trade under end labels (our calculation, based on the unit conversions given in the paper: daily to per-trade). Secondly, the signal is a “relative return” (stock minus index with a coefficient of 1), while the position is long in the stock only, so the market movement over the holding window is included in the profit and loss of the position. All five stocks are large individual items in the index to which they belong, so the relative return is to some extent mechanical, and the authors should state the weight of the five-stock index as an entire. Last, the 1.0 and 1.5 times the sample standard deviation of the 15-minute returns are what’s called sets of ‘barrier multipliers’ and are not theoretically linked to the 15-minute time horizon or the reward-to-risk ratio.

Questions for the authors. Does the signal have a positive gross return compared to entering at random times on the same stocks? Why is the beta of the stock to the index set to one, and how much does the signal change if the beta is misestimated? Which of the predictions from the stop-loss theory is being tested and which of the results would be a counterexample?

3. Concepts (Weaknesses, Strengths or Gaps)

There are several concepts that are well explained. The separation of independent-session accounting from funded accounting, the definition of expected shortfall as fractional weights, and the distinction between an abstention, a valid no-trade day, an unresolved outcome and a funding failure (p. 5) all decrease the ambiguity.

Other ideas are not as well developed. The term robustness of a path is never precisely defined. In the following tables there are three sample definitions: 443, 360 or 381 and 443, and the reader is asked to figure out why each is applicable. A nominal 95% interval is reported, but not then given an inferential interpretation and the purpose of the interval is not clear. The funded drawdown in Table 9 is also tricky to use as a measure of risk. The time-only mean continues to lose session by session, with a total loss of approximately 10 per cent over 443 sessions equal to minus 2.34 basis points per session, which is of the same magnitude as the reported maximum drawdowns of between 9.4 and 10.4 per cent. This is our approximate calculation, but it suggests that the ordering of the drawdowns is primarily about ordering by mean return and not about independent information about path risk.

The other conceptual issue is the unresolved timestamp convention (p. 3). Both interpretations are carried pending supplier confirmation, which doubles every table. In many minute files the convention can be deduced from the data (e.g. by checking for the existence of a bar with a label of 09:15 and a bar with a label of 15:30); the authors should give the appropriate indication in this case. Also, there is no obvious reason why the two interpretations should lead to the two different numbers of evaluable sessions: 360 and 381, corresponding to a difference of 21 sessions, too large for a one-minute relabelling.

Questions for the authors. Is it possible to determine the timestamp convention using the bar inventory? Is the drawdown in Table 9 informative once the negative drift is removed, for example by detrending? What decision would a practitioner take differently if an interval were exploratory as opposed to confirmatory?

4. Contribution or Extension of the Topic

The main noticeable contribution of the paper is reproducible auditing. The matched ledger design, the frozen protocol with hashes, the lack of backfilling for missing data, and the synthetic coverage experiment with an autoregressive coefficient of 0.7, showing that the five-day block bootstrap undercovers (78.5 and 81.5 percent), are valuable and honestly presented. In addition, the suggestion that the same audit be applied to broker or platform backtests based on an undisclosed candle sequence (p. 15) is a useful one.

The substantive extension of the exit-policy literature is limited. The study employs just one entry rule, five highly liquid stocks, a long-only strategy and a 15-minute time frame, and the reported effect is extremely modest. A difference of 0.2325 basis points of C0 per session would translate to INR 2.3 per day on INR 100,000, using the authors' own method of converting basis points to INRs (p. 9). Even a statistically reliable difference of this size would have little economic content, so the contribution is related more to the method than the finding.

Questions for the authors. What economic magnitude would justify a recommendation to use or avoid a profit target? Is it possible to broaden the cross-section (for instance to the complete index) in order to get more episodes per day with the 15-minute horizon and to improve the power of the comparison? Is the claim of novelty over Kaminski and Lo and over Lo and Remorov stated precisely enough to be tested?

5. Methodology (Strengths or Gaps)

Strengths. The signal incorporates only past information, the universe of stocks and the holding rule are static and pre-specified before considering outcomes, positions are matched across policies, the exclusions are clearly specified, and the split is chronological with the authors specifying which procedures were added after the first development run (p. 13). The reconciliation of 84 policy means to a very small tolerance is a good practice as are the 104 software tests (p. 10). The paired contrasts in Table 6 are internally consistent with the difference between the policy means, as are the numbers of 60, 20 and 20 percent of 739

sessions, and the final partition of 148 sessions, with 9.000 and 9.525 days equal to 2.5 percent of 360 and 381, respectively; the Monte Carlo standard errors in Table 10 are consistent with the binomial formula for 200 replications; and the ratios of mean width to oracle width of 0.965 and 0.978 are consistent with the width of the oracle being 3.92 divided by the square root of the number of sessions for unit variance.

We also reproduced the pilot. Tables 5, 6 and 7 match the supplied scenario means, paired contrasts and sensitivity files; the independent reconciliation file agrees with the counts in the text (57 signals, 2,394 executions, 1,680 accounts, 399 episodes, 14 runs and 854 ledger records); the reported memory use agrees with the resource file; and the synthetic both-hit fixture returns minus 150 and plus 50 basis points under the two candle orders, as stated. The notebook implements the same past-only signal, the same segment rule for bootstrap blocks, and a freeze check that prevents the final test if code, data and protocol hashes fail to match, and it marks all intervals as exploratory. The reproducibility statements are thus well supported and the rest of our comments are of a design and interpretation nature.

Gaps: There are eight problems that hinder interpretation.

(a) Power. The study has low power and does not say so quantitatively. The width of the interval for the comparison of the start-label is approximately 0.52 basis points, corresponding to a standard error of approximately 0.13 and a ratio of the point estimate to the standard error of approximately 1.8, before the documented undercoverage. With standard power, this approximation allows the design to detect only differences greater than approximately 0.37 basis points per session. The position is weaker with 148 final-test sessions, and, according to the authors, only 3.70 days (p. 7) are in the final expected-shortfall tail. A brief description of the minimal effects that would be detectable by the final test would facilitate the reader’s consideration of the value of a completed test.

(b) Use of the validation block. No parameter is estimated, so the validation block does not validate a model, and the authors say that the block will be used primarily to detect bugs and errors in the metadata (p. 6). The confirmatory sample is therefore reduced to 148 sessions as 20% of the calendar is set aside for this purpose. If the protocol can be frozen after the development stage, validation and final sessions could be combined into one confirmatory sample of about 296 sessions, doubling the evidence without compromising the design process.

(c) Cost and fill assumptions. All policies have the same cost model, so that the cost scenarios in Table 7 give the same contrasts (the rows of 5 basis points and 25 basis points repeat the reference row). The central economic issue with exits is that a stop is placed as a market order, which crosses the spread, while a target is a resting limit order that can be filled or not; therefore there should be variation in exit-specific costs and fill probabilities. The present assumptions also treat a touch of the target as a fill, and the engine fills a gap above the target at the target price, not at the open price, while a gap through the stop is filled at the open price. This was verified in the code and in the synthetic cases (when an opening target gap is created, it gets filled at the target, and when an opening stop gap is

created, it gets filled below the stop). Both rules are unfavorable to the combined policy, so the design is conservative against a profit target, and the paper ought to make it clear that this is the case.

(d) Designing barriers and the ambiguity audit. The two distances of the stop and target are set at one and one and a half sample standard deviations and at these distances a single minute bar is very unlikely to cross both barriers, so that a zero ambiguity count is close to mechanical. A report of a full grid of barrier multiples, not selected for performance, would indicate where ambiguity of the candles starts to become an issue and whether or not conclusions are reliable. This would also provide the title with its rationale.

(e) Signal stability. There are 33 unique episodes under start labels and 24 under end labels in the pilot (Table 5), a difference of approximately 27 percent when relabelled by one minute. This means that a z-score calculated over 20 sessions is a choppy indicator. The distribution of signal counts should be reported as robustness rather than as optimisation and so should be the sensitivity to window length and threshold.

(f) Inference. The Monte Carlo standard error of a five-day block bootstrap, with 200 synthetic replications per design, is two to three percentage points, and is too coarse to calibrate a 95 percent interval. The abstract’s assertion that undercoverage occurs under more severe serial dependence is not quite correct because of the start-label independent case, which has a coverage of 90.5 percent with a standard error of 2.07, or more than two standard errors below 95. The diagnostic also simulates Gaussian differences, a limit that the notebook itself acknowledges. We recommend at least 1,000 replications, a comparison with the stationary bootstrap of Politis and Romano, a sign-flip permutation test for the paired differences, and an analysis at the level of episodes clustered by date to make use of the 637 and 662 episodes.

(g) Missing-data rule. The rule that a missing day removes the following 20 sessions from the evaluable set is conservative but expensive: the pilot loses about half of its evaluation days, and the development run loses 83 or 62 of 443 sessions, depending on which labeling convention is used. Given the estimand involves the complete case sessions, the authors should report if there are differences between the volatility of the excluded sessions, and/or if there are differences in the signal frequency of the excluded sessions, and should consider recomputing the rolling window over only the available sessions as a sensitivity check.

(h) Multiple scenarios. Fourteen scenarios are assessed, without any multiplicity corrections. The one main comparison is rightly identified, but the sensitivity results should be described as descriptive, and the literature on data-snooping beyond Sullivan, Timmermann and White (for example Hansen; Bailey et al.) could be used as the basis for describing the protocol.

Questions for the authors. The appendix gives Python 3.13.15, the pilot manifest 3.12.14; which environment did the development tables come from? What is the smallest difference that the final test can detect? Why not pool the validation and final sessions? What are the results if stops are represented as market orders, where the spread cost is added, and

targets as limit orders, where the fill probability is less than one? Over what grid of barrier multiples does the ambiguity count remain zero?

6. Literature Review (Weaknesses or Gaps)

The review is focused and contains sources on reversal and costs, stop-loss theory, candle ambiguity, data snooping and estimation of tail-risk. It also identifies a current limitation for every source and connects it to that limitation, which is a plus. Several gaps remain. The short-term reversal evidence that motivates the entry rule is only captured in the work of Heston, Korajczyk and Sadka and de Groot, Huij and Zhou, but not by Jegadeesh. The residual-reversion method with standardised score thresholds is treated by Avellaneda and Lee but is not cited, although it is the most similar trading method to today's entry rule. The evidence by Gao et al. on intraday momentum and market level reversals is missing and the proposal fails to consider that spreads might be estimated based on high and low prices (Corwin and Schultz). The treatment of data snooping relies on a single discussion paper from 1998, while the standard tools are the Reality Check, the test for superior predictive ability and the probability of backtest overfitting. Lastly, there is no mention of the research conducted on Indian intraday markets except one dissertation on pairs trading, and the authors need to look for such literature because the setting of the work is the NSE microstructure.

Other citations require tightening. Bollerslev and Siggaard are mentioned in the text, but "Aleti, Saketh, et al." is listed in the reference list on p. 16. A few sentences that summarise sources are hard to link up, and without the originals it is impossible to know which source they are summarising, for example the statement that the return to a stop overlay "is not possible to increase the expected return" (p. 2) and the sentence on Heston, Korajczyk and Sadka at the beginning of the Theoretical Framework. We suggest that each be restated as a specific claim and its conditions.

Questions for the authors. What is the connection between the entry rule and the classical short-term reversal and the statistical arbitrage literature? Which of the suggested data-snooping methods are applicable to 5 stocks and which are not?

7. Style

The proposal is well structured, the notes provided under each table are informative, and the tables are easy to read. The writing is frequently obscure, however, and this detracts from the care with which the content is provided. Examples include the circular sentence "So, we compare the exits with the same entry rule to compare them" (p. 1); the headings of "Rational and limitations of fixed design" (p. 4), "The amount estimated and accounting" (p. 5), "In two observation frequencies there is funded equity" (p. 12) and "Calculating and interpreting bootstraps" (p. 13); and the table captions in the same way as sentence fragments, such as "Table 8. In the reference execution, an expanded development comparison is performed." There are also contradictions in the tenses within paragraphs and many abbreviations (ES97.5, ADB, C0, O-L-H-C) are used before they are explained. There are

several passages that appear to be translated or machine-assisted and a careful editing by a fluent reader would eliminate ambiguity in the following abstract: “The mean returns are lower for combined exits than time-only exits”.

It is long for a proposal (19 pages) and a significant portion of the Results section specifies provenance such as commit identifiers, hashes, manifest names, which is better suited for an appendix. Table 1 is written for the instructor, and may be omitted from the text, as it responds to previous feedback from the course. There is a problem with the overall formatting of the reference list: the URLs have replacement errors (“nSEindia” for “nseindia”), National Stock Exchange has been included both before and after “The”, and the punctuation is inconsistent (“Lo, Andrew W.; and Alexander Remorov” and “et al.,” at the end of the Sullivan entry).

Questions for the authors. Does each section begin with the purpose of the section in one sentence and end with the implication of the section to the topic question? Is it possible to rearrange the abstract to be Motivation, Method, Finding and Limitation?

8. Results and Conclusions (Weaknesses or Gaps)

The results reported are tentative, and, where possible, internally consistent. Combined exits earn lower mean returns than time-only exits (minus 0.2325 and minus 0.1273 basis points), have lower expected shortfall (14.41 against 17.41 basis points under start labels) and larger funded drawdown, and the intervals include zero. The conclusion that the results “do not demonstrate overall superiority” is acceptable (p. 15).

There are 4 points that need to be addressed. First, the above statement about the increased development comparison “does not create a mean return advantage” is more powerful than the evidence, as a non-significant interval is not evidence of no effect. The start label interval is approximately -0.44 to +0.08 basis points, which fits a modest harm as well as a modest gain, and the conclusion should be that “the estimate is negative but imprecise.” Second, the sensitivity numbers given in the Discussion (0.6889 and 0.6430 for 10 basis points of stop slippage, and 0.1174 and 0.0722 for 5 minutes of delay, p. 14) cannot be located in a table for the development sample. The pilot and development samples are different, so a delay effect does not need to agree, however, Figure 1 shows points only, and a numeric table similar to Table 7 with the number of sessions aligned in each row is necessary. Third, the negative drift (as noted above) complicates the funded drawdown conclusion, while the expected-shortfall conclusion is based on 9.000 and 9.525 tail days, so the leave-one-day-out ranges should be presented in the paper, rather than just in the output files. Fourth, if there is no episode that is ambiguous, the agreement of the two candle paths on the reference sample is not informative, and the conclusion should not be given as the sign of robustness.

Questions for the authors. Is it possible to restate the conclusion, for example as an estimated return reduction of up to about 0.4 basis points per session, with a possible slight increase? Where are the sensitivity numbers for the development sample, and do they agree

with the numbers shown in Figure 1? What is the ranking of the policies if the single worst day is removed?

Summary of Priority Recommendations

1. Align the title and abstract with the actual finding, and state the research question in the abstract.

2. Report the gross return of the entry signal against a random-entry baseline, and address the mismatch between the relative signal and the absolute long position.

3. State the minimum detectable effect, pool validation and final sessions if the protocol permits, and consider a wider cross-section to increase power.

4. Replace the policy-invariant cost scenarios with exit-specific costs and fill probabilities, and report a full grid of barrier multiples with the ambiguity count for each.

5. Strengthen the inferential diagnostics (more replications, stationary bootstrap, permutation test) and rewrite the conclusion in terms of interval and effect size.

6. Add a numeric sensitivity table for the development sample, make drawdown comparisons robust to drift, and correct the reference list and the language of the headings and captions.

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